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QUICK-RELEASE WEIGHT PIN FOR WEIGHT STACK ASSEMBLY AND A WEIGHT STACK ASSEMBLY INCORPORATING THE SAME

Abstract

The invention discloses a quick-release weight pin for weight stack assembly and a weight stack assembly incorporating the same, comprising: a sleeve, a pin rod, a first spring, and a locking assembly, wherein one end of the sleeve is provided with an internally communicating through-hole. The pin rod and the sleeve are coaxially disposed, wherein the pin rod disposed within the through-hole of the sleeve and configured to perform axial extendable and retractable movement. The first spring is disposed within the sleeve, wherein the first spring is mounted around the pin rod with both ends respectively abutting against the pin rod and the sleeve.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to the technical field of fitness equipment accessories, and more particularly to a quick-release weight pin for weight stack assembly and a weight stack assembly incorporating the same.

BACKGROUND ART

[0002] In existing fitness equipment, weight stack assembly is common components used for resistance training. When utilizing weight stack assembly for training exercises, current implementations require setting multiple distinct weight levels. However, the critical technical challenge for practical training applications lies in achieving automatic weight adjustment of the weight stack assembly without interrupting ongoing training operations.

[0003] At present, disclosed solutions comprise utilizing weight pin to adjust weight stack assembly; however, existing weight pin require verification of secure locked state when inserted into weight stack assembly. Consequently, most implementations necessitate simultaneous two-handed operation to achieve secure locking, presenting operational challenges for users.

INVENTION CONTENT

[0004] To overcome the said drawbacks, an objective of the present invention is to provide a quick-release weight pin for weight stack assembly that ensures the locking portion and the pin main body maintain relatively stable fixed states. Consequently, during use, the weight pin can be inserted into the weight stack assembly through single-handed operation without requiring special attention, thereby providing operational convenience.

[0005] To achieve the above objectives, the technical solution adopted by the present invention is: a quick-release weight pin for weight stack assembly, comprising: [0006] A sleeve having a through-hole communicating with its interior disposed at one end; [0007] A pin rod coaxially disposed with said sleeve, said pin rod extending through said through-hole and being axially extendable and retractable; [0008] A first spring positioned within said sleeve, said first spring being sleeved over said pin rod with both ends respectively abutting against said pin rod and sleeve; [0009] A locking assembly comprising a locking lever, one end of said locking lever being pivotally connected to said sleeve and the other end forming a hand-held end, said locking lever having a locking portion on its sleeve-facing side configured to extend into said sleeve and abut against said pin rod; [0010] Wherein when said locking portion abuts against said pin rod, said first spring remains in a compressed state and said locking assembly maintains static relative to said sleeve.

[0011] The advantageous effects of the quick-release weight pin for weight stack assembly according to the present invention are as follows: [0012] {circle around (1)} When the pin assembly of the present application is to be used to a weight stack assembly, operation only requires engaging the locking portion in abutting against the pin rod, wherein the locking assembly remaining stationary relative to the sleeve, thereby enabling single-handed operation by the user for effortless insertion into the weight stack assembly. [0013] {circle around (2)} In the sleeve, the provision of a through-hole restricts the movement direction of the pin rod; in the locking assembly, a locking lever is pivotally connected at one end to the sleeve to enable the locking lever to pivoting motion along the sleeve, thereby driving the locking portion to enter or exit the sleeve, achieving locking or unlocking of the pin rod. [0014] {circle around (3)} The opposite end of the locking lever is configured as a hand-held end, such that depressing the locking lever is mechanically leveraged for reduced exertion. By virtue of the locking portion's engagement with the pin rod, the pin rod is stably inserted into the weight stack assembly to maintain weight stability during exercise. When weight adjustment is required, the locking portion is rapidly disengaged through pivotal motion of the locking lever, and the pin rod is promptly retracted via restorative

force from a first spring, thereby enabling efficient removal of the quick-release weight pin for weight stack assembly from the weight stack assembly.

[0015] More specifically, when said locking portion abuts against said pin rod, said locking lever is oriented parallel to said pin rod, with said locking portion disposed perpendicular to said pin rod. Given that the pin rod is axially extendable/retractable, the perpendicular orientation of the locking portion relative to the pin rod generates an axial blocking force, thereby restraining axial displacement of the pin rod.

[0016] Furthermore, the end of said locking portion distal from said locking lever is provided with an arc-shaped chamfer, configured to mitigate frictional wear on the pin rod surface caused by contact with the locking portion during operational motion.

[0017] Furthermore, a second spring is comprised, said second spring is disposed within said sleeve, having one end abutting within said sleeve and the opposite end oriented toward said pin rod. When said first spring is in an uncompressed state, said second spring remains positioned abutting against the pin rod. The arrangement of the second spring prevents the pin rod from impacting the inner wall of the sleeve due to excessive restorative force from the first spring during unlocking, thereby providing controlled buffering.

[0018] Furthermore, an inner wall of said sleeve at an end distal from the through-hole is provided with a retention seat configured to receive said second spring. The retention seat constrains axial and radial displacement of the second spring.

[0019] Furthermore, said sleeve is disposed with an access hole in its sidewall that communicates with the interior of the sleeve. Said locking portion is configured to extend into said sleeve through said access hole.

[0020] Furthermore, a first pivot seat is disposed on one side of said access hole, and a second pivot seat corresponding to said first pivot seat is disposed with the side of said locking lever distal from said hand-held end. Said first pivot seat and said second pivot seat are pivotally secured by a shaft.

[0021] Furthermore, when said locking portion is abutted against said pin rod, a clearance is defined between said locking portion and one end of said access hole that is distal from said pivot seat. During the pivoting movement of the locking lever, the locking portion undergoes an axial displacement along the pin rod. The provision of the clearance ensures that the locking portion can smoothly enter into the sleeve.

[0022] Furthermore, said pin rod comprises a rod body and an end cap arranged coaxially with said rod body, wherein said end cap has a radial dimension greater than that of said rod body, and wherein one end of said end cap forms a first end surface against which said first spring is configured to abut, and another end of said end cap forms a second end surface against which said locking portion is configured to abut.

[0023] The present invention further provides a weight stack assembly capable of quick-release weight, comprising a plurality of weight plates and a shuttle shaft movably disposed through each of said weight plates. The weight stack assembly further comprises said quick-release weight pin for weight stack assembly. A fixation hole configured to receive the penetration of pin rod of said quick-release weight pin is provided on said shuttle shaft, and a connection hole configured to receive the insertion of said quick-release weight pin is provided on each of said weight plates.

[0024] Furthermore, when said first spring is in a compressed state, said pin rod and said hand-held end are configured to be inserted into connection hole of two adjacent weight plates, respectively.

Description

DESCRIPTION OF THE DRAWINGS

[0025] FIG. 1 IS A PERSPECTIVE VIEW ILLUSTRATING A LOCKED STATE OF A QUICK-

RELEASE WEIGHT PIN FOR WEIGHT STACK ASSEMBLY IN ACCORDANCE WITH EMBODIMENT 1 OF THE PRESENT INVENTION;

[0026] FIG. 2 IS A SIDE VIEW ILLUSTRATING THE LOCKED STATE OF THE QUICK-RELEASE WEIGHT PIN FOR WEIGHT STACK ASSEMBLY IN ACCORDANCE WITH EMBODIMENT 1 OF THE PRESENT INVENTION;

[0027] FIG. 3 IS A CROSS-SECTIONAL VIEW TAKEN ALONG LINE A-A IN FIG. 2;

[0028] FIG. 4 IS A CROSS-SECTIONAL VIEW ILLUSTRATING AN UNLOCKED STATE OF A QUICK-RELEASE WEIGHT PIN FOR WEIGHT STACK ASSEMBLY IN ACCORDANCE WITH EMBODIMENT 1 OF THE PRESENT INVENTION;

[0029] FIG. 5 IS A CROSS-SECTIONAL VIEW ILLUSTRATING THE WEIGHT PIN ENGAGED WITH A WEIGHT PLATE ACCORDING TO EMBODIMENT 2 OF THE PRESENT INVENTION;

[0030] FIG. 6 IS A CROSS-SECTIONAL VIEW ILLUSTRATING THE HAND-HELD END BEING PUSHED ASIDE BY THE NEXT WEIGHT PLATE DURING A WEIGHT REDUCTION TRAINING EXERCISE IN ACCORDANCE WITH EMBODIMENT 2 OF THE PRESENT INVENTION.

IN THE DRAWINGS

[0031] 1—SLEEVE; 11—RETENTION SEAT; 12—ACCESS HOLE; [0032] 2—PIN ROD; 21—ROD BODY; 22—END CAP; [0033] 3—FIRST SPRING; [0034] 4—LOCKING ASSEMBLY; 41—LOCKING LEVER; 411—HAND-HELD END; 42—LOCKING PORTION; 421—ARC-SHAPED CHAMFER; [0035] 5—SECOND SPRING; 61—WEIGHT PLATE; 611—CONNECTION HOLE; 62—SHUTTLE SHAFT.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0036] Preferred embodiments of the present invention are described in detail below with reference to the accompanying drawings, to enable those skilled in the art to more clearly comprehend the advantages and features of the present invention, thereby providing clearer definition of its protection scope.

Embodiment 1

[0037] Referring to FIGS. 1-3, the present embodiment discloses a quick-release weight pin for weight stack assembly. The quick-release weight pin for weight stack assembly of this embodiment comprises: sleeve 1, pin rod 2, first spring 3, and locking assembly 4, wherein the sleeve 1 comprises a through-hole in communication with its interior at one end. The pin rod 2 is coaxially disposed with the sleeve 1, disposed through the through-hole, and configured for axial extendable and retractable movement. The first spring 3 is positioned within the sleeve 1, concentrically mounted on the pin rod 2, with opposing ends abutting against the pin rod 2 and the sleeve 1, respectively. The locking assembly 4 is disposed externally on the sleeve 1 and comprises a locking lever 41. One end of the locking lever 41 is pivotally connected to the sleeve 1, while another end thereof forms a hand-held end 411. A sleeve-facing side of the locking lever 41 is disposed with a locking portion 42 configured to extend into the sleeve 1 and operatively abut against the pin rod 2. When the locking portion 42 abuts against the pin rod 2, the first spring 3 is in a compressed state, wherein the locking assembly 4 maintains static engagement relative to the sleeve 1.

[0038] In an initial state, the locking portion 42 disengages from the pin rod 2, the first spring 3 returns to a neutral uncompressed condition, and the pin rod 2 attains a minimal extension length from the sleeve 1. To insert the quick-release weight pin for weight stack assembly into a weight stack assembly, the pin rod 2 is actuated to extend axially through the through-hole to a predetermined engagement position, thereby compressing the first spring 3. Subsequently, the hand-held end 411 is depressed to drive the locking portion 42 into the sleeve 1.

[0039] When locking the position of the pin rod 2, first apply pressure to the pin rod 2 toward one end of first spring 3, then rotate hand-held end 411 toward the end adjacent to sleeve 1. If locking portion 42 contacts the end of pin rod 2 distal from first spring 3 during its extension into sleeve 1,

release pin rod 2. Through the abutment effect of locking portion 42, pin rod 2 maintains static position against the resilient restoring force of first spring 3, until locking portion 42 reaches a predetermined locking position. At this stage, locking portion 42 remains stationary through cooperative action between first spring 3 and locking portion 42 itself, maintaining constant extension length from sleeve 1. At this stage, the extended end of the pin rod 2 projecting from sleeve 1 may be inserted into a weight stack assembly to secure said weight stack assembly, wherein through the locking engagement of locking portion 42 with pin rod 2, axial retraction of pin rod 2 during user exercise is prevented. Consequently, the pin rod 2 maintains stable positioning within the weight stack assembly, ensuring stable weight configuration.

[0040] If locking portion 42 does not contact pin rod 2 during its extension into sleeve 1, said locking portion 42 is directly advanced to the predetermined locking position. At this phase, pin rod 2 may be released to axially retract under the resilient restoring force of first spring 3 until abutting against locking portion 42, whereby the abutment interaction between locking portion 42 and pin rod 2 ensures positional stability of pin rod 2. The extended end of pin rod 2 projecting from sleeve 1 is then inserted into a weight stack assembly to fixedly secure said weight stack assembly, wherein through locking engagement of locking portion 42 with pin rod 2, axial retraction of pin rod 2 during exerciser motion is prevented, whereby pin rod 2 maintains stabilized positioning within the weight stack assembly to ensure weight configuration stability.

[0041] In the present application, locking portion 42 is configured to extend into sleeve 1 (thereby providing increased contact area in the radial direction of pin rod 2) and abut against the end portion of pin rod 2, wherein pin rod 2 is biased under the compressive force from first spring 3 during this operational phase, such that locking assembly 4 maintains a relatively stable static state with respect to sleeve 1. During usage, the risk of accidental dislodgement of locking lever 41 is eliminated, thereby enabling single-handed operation of the quick-release weight pin when engaged with weight stack assembly, thus providing enhanced operational convenience.

[0042] It is specified that during extraction of pin rod 2, the end of pin rod 2 within sleeve 1 must be positioned to the right side of locking portion 42 at its locking position (as shown in FIG. 3), thereby ensuring proper locking engagement of locking portion 42 with pin rod 2.

[0043] To ensure stable locking engagement of locking portion 42 with pin rod 2, in some embodiments (as shown in FIG. 3), when locking portion 42 abuts against pin rod 2, locking lever 41 is disposed parallel to pin rod 2 while locking portion 42 extends perpendicular to pin rod 2, such that a lateral surface of locking portion 42 achieves full surface contact with an end portion of pin rod 2. As pin rod 2 is axially extendable and retractable, when locking portion 42 is oriented perpendicular to pin rod 2, it provides an axial blocking force to pin rod 2, thereby preventing axial displacement of pin rod 2.

[0044] Furthermore, during pivotal motion of locking lever 41, locking portion 42 transitions from an inclined orientation to a perpendicular orientation relative to pin rod 2. Should locking portion 42 abuts against pin rod 2 during this transitional motion, frictional wear to pin rod 2 may potentially occur. Furthermore, in some embodiments, an arc-shaped chamfer 421 is disposed at the end of the locking portion 42 distal from the locking lever 41, so as to minimize frictional wear on the surface of the pin rod 2 caused by contact with the locking portion 42 during movement.

[0045] During unlocking, the pin rod 2 is retracted into the sleeve 1 under the action of the first spring 3. To prevent the pin rod 2 from impacting against the inner wall of the sleeve 1 due to excessive restorative force of the first spring 3, in some embodiments, a second spring 5 is additionally disposed within the sleeve 1. The second spring 5 is disposed along the axial direction of the sleeve 1, with one end abutting against the sleeve 1 and the other end oriented toward the pin rod 2. As shown in FIG. 4, when the first spring 3 is in its natural state, the second spring 5 is capable of abutting against the pin rod 2. When the pin rod 2 is in a locked state, the end of the second spring 5 facing toward the pin rod 2 remains in a unsupported state.

[0046] Furthermore, to positionally constrain the second spring 5, a retention seat 11 is disposed on

the inner wall of the sleeve **1** at an end distal from the through-hole, configured to have the second spring **5** sleeved thereover.

[0047] In some embodiments, as shown in FIGS. **3-4**, an access hole **12** is disposed on the lateral wall of sleeve **1** in communication with its interior, through which the locking portion **42** is configured to enter into the sleeve **1**. Furthermore, a first pivot seat is disposed at one end of access hole **12**, and a second pivot seat correspondingly disposed with the first pivot seat is disposed at an end of the locking lever **41** distal from the hand-held end. The first pivot seat and the second pivot seat are pivotally connected and secured via a shaft.

[0048] During pivoting motion of the locking lever **41**, the locking portion **42** undergoes axial displacement along the pin rod **2**. To ensure smooth insertion of the locking portion **42** into the sleeve **1**, dimensional parameters of the access hole **12** are required to be defined. Specifically, when the locking portion **42** abuts against the pin rod **2**, a designed clearance is maintained between the locking portion **42** and an end of the access hole **12** distal from the pivot seat. Such designed clearance configuration ensures that the access hole **12** provides sufficient allowance to enable unobstructed ingress of the locking portion **42**.

[0049] When unlocking the pin rod **2**, the locking lever **41** is required to be pivoted upwardly to withdraw the locking portion **42** from the sleeve **1**. When the locking portion **42** abuts against the pin rod **2**, the hand-held end **411** is configured to extend outwardly from the sleeve **1**, thereby enabling application of an upward actuation force to the hand-held end **411** during unlocking operations.

[0050] In some embodiments, as shown in FIG. **4**, the pin rod **2** comprises a rod body **21** and an end cap **22** coaxially disposed therewith. The end cap **22** has a radial cross-sectional dimension greater than that of the rod body **21**. A first end surface of the end cap **22** is configured to abut against the first spring **3**, while a second end surface thereof is configured to engage with the locking portion **41** as a abutment interface.

Embodiment 2

[0051] As shown in FIGS. **5-6**, this embodiment provides weight stack assembly capable of quick-release weight, comprising a plurality of weight plates **61** and a shuttle shaft **62** movably disposed through each of the weight plates **61**, the present embodiment comprises a quick-release weight pin for weight stack assembly in accordance with embodiment 1. The shuttle shaft **62** is formed with a fixation hole configured to receive insertion of the pin rod of the quick-release weight pin, and each weight plate is provided with a connection hole **611** configured to allow extension of the quick-release weight pin therethrough.

[0052] Prior to training, the weight configuration pin of embodiment 1 is initially set to a locked state. A plurality of weight pins (the quantity determined by desired weight gradations) are inserted into weight plates **61** positioned at different elevations (wherein the pin rod **2** and hand-held end **411** of each weight pin are operatively disposed within connection holes **611** adjacent to upper and lower weight plates **61** respectively), thereby corresponding to the different configuration weights. Subsequently, the shuttle shaft **62** is pulled upwardly, wherein the weight plate **61** positioned below the lowermost weight pin is caused to ascend with said weight pin, thereby extracting the hand-held end **411** from the weight plate **61**, while the hand-held end **411** progressively deploys into a non-interfering configuration with the ascending upper weight configuration plate **61** (as shown in FIG. **6**). Wherein respective weight plates **61** disposed below said weight pin remain stationary during said upward pulling operation.

[0053] At this operational stage, while the locking assembly **4** ceases to exert thrust force on the pin rod **2**, said pin rod **2** is maintained in a retracted configuration under compressive loading from both the shuttle shaft **62** and weight plate **61**; during descent of the weight plates **61**, the shuttle shaft **62** and weight plates **61** discontinue application of said compressive loading to the pin rod **2**, whereby the pin rod **2** is automatically protruded to an unlocked position by biasing force of the first spring **3**.

[0054] Upon successive upward pulling of the weight plates **61**, the weight plate **61** disposed below a superiorly positioned weight pin operatively continues to exert a downward pivoting force on the hand-held end **411** associated with said weight pin, thereby transitioning said hand-held end **411** into the configuration substantially as shown in FIG. **6**. Through iterative repetition of said disengagement process between the weight pin and corresponding weight plate **61**, a self-regulated operational cycle is established to facilitate expeditious weight selection across multiple resistance levels, wherein the entire weight adjustment sequence operates autonomously without necessitating manual intervention.

[0055] The foregoing embodiments are provided solely to exemplify the inventive technical concepts and distinctive features of the present invention, said embodiments being configured to enable persons skilled in the art to comprehend and practice the disclosed innovations, without being construed as limiting the claimed scope of protection, wherein all equivalent variations and modifications embodying the inventive essence shall fall within the purview of the appended claims and their legal equivalents.

Claims

1. A quick-release weight pin for weight stack assembly, characterized in that it comprises: A sleeve having a through-hole communicating with its interior disposed at one end; A pin rod coaxially disposed with said sleeve, said pin rod extending through said through-hole and being axially extendable and retractable; A first spring positioned within said sleeve, said first spring being sleeved over said pin rod with both ends respectively abutting against said pin rod and sleeve; A locking assembly comprising a locking lever, one end of said locking lever being pivotally connected to said sleeve and the other end forming a hand-held end, said locking lever having a locking portion on its sleeve-facing side configured to extend into said sleeve and abut against said pin rod; Wherein when said locking portion abuts against said pin rod, said first spring remains in a compressed state and said locking assembly maintains static relative to said sleeve.
2. The quick-release weight pin for weight stack assembly according to CLAIM 1, characterized in that: wherein when said locking portion abuts against said pin rod, said locking lever is parallel to said pin rod, said locking portion is perpendicular to said pin rod; and said pin rod and hand-held end are respectively insertable into adjacent weight stack assembly.
3. The quick-release weight pin for weight stack assembly according to CLAIM 1, characterized in that: said locking portion has an arc-shaped chamfer at its distal end away from said locking lever.
4. The quick-release weight pin for weight stack assembly according to CLAIM 1, characterized in that further comprising a second spring: said second spring positioned within said sleeve, having a first end abutting against an interior of said sleeve, a second end oriented toward said pin rod; wherein when said first spring is in a natural state, said second spring is abutting against said pin rod.
5. The quick-release weight pin for weight stack assembly according to CLAIM 4, characterized in that: an inner wall of said sleeve at an end distal from said through-hole is formed with a retention seat for said second spring set thereover.
6. The quick-release weight pin for weight stack assembly according to CLAIM 1, characterized in that: said sleeve has a lateral wall formed with an access hole communicating with its interior; said locking portion being configured to enter said sleeve through said access hole.
7. The quick-release weight pin for weight stack assembly according to CLAIM 6, characterized in that: a first pivot seat is disposed at one end of said access hole, a second pivot seat corresponding to said first pivot seat is disposed at an end of said locking lever distal from said hand-held end; said first and second pivot seats being pivotally connected by a shaft.
8. The quick-release weight pin for weight stack assembly according to CLAIM 1, characterized in that: said pin rod comprises a rod body and an end cap coaxially disposed with, wherein said end

cap has a radial dimension greater than that of said rod body, and wherein said end cap is formed with a first end surface on one side thereof configured to abut against said first spring, and a second end surface on an opposing side thereof configured to be abutted against the locking portion. Therefore, by providing the end cap structure, the locking portion can exert a more stable abutting and pushing effect on the pin rod through the end cap, while ensuring stable abutment between the first spring and the pin rod.

9. A weight stack assembly capable of quick-release weight comprising a plurality of weight plates (61) and a movable shuttle shaft (62) penetrating through each said weight plate (61), characterized in that: the weight stack assembly comprises a quick-release weight pin for weight stack assembly as said in CLAIM 1, wherein said shuttle shaft (62) is provided with a fixation hole configured for said pin rod (2) of said quick-release weight pin to pass through, and each said weight plate (61) is provided with a connection hole (611) configured for the quick-release weight pin to engage with.

10. The weight stack assembly capable of quick-release weight according to CLAIM 9, characterized in that: wherein when said first spring (3) is in a compressed state, said pin rod (2) and said hand-held end (411) are configured to extend into connection holes (611) of two adjacent weight plates (61), respectively.
